IndiGo Profitability Intelligence

D4-v1 | CO3 | 07 November 2026

Next-quarter PAT margin forecasting using quarterly financial and macro/market indicators.
35 supervised observations | Target range: 2017Q4 to 2026Q2
Best validation MAE: 12.18 percentage points (persistence baseline)
Locked test MAE: 11.53 percentage points (persistence baseline)

1. Problem type and model choice

Business question: can historical IndiGo profitability and lagged currency, inflation and Brent benchmark indicators support a one-quarter-ahead PAT-margin forecast?
Target: PAT / Revenue from Operations x 100. This is supervised regression in percentage points, not a stock-price prediction.
Source: data/processed/Final_Quarterly_Data(1).xlsx, sheet Final_Quarterly_Data.
Thirty-eight rows are present; two rows marked Not available in source are excluded.
One-quarter lagging leaves 35 adjacent reported-quarter examples.

Why these model families fit

Persistence baseline: carry forward the last reported PAT margin. Transparent time-series
benchmark, particularly appropriate for a small quarterly sample.
Ridge regression: regularised linear challenger; scaling and L2 shrinkage help with correlated macro indicators and limited observations.
Random Forest regression: nonlinear challenger for possible thresholds/interactions; shallow trees and larger leaves constrain overfitting.
D4 decision: neither candidate beats persistence on validation; retain the baseline.

Features and validation design

2. Declared time-aware split

The split windows were fixed before tuning, are contiguous/disjoint and are constants in src/d4_train.py. No random shuffle is used.
TRAIN 2017Q4-2022Q4 21 rows Fit model and preprocessing.
VALIDATION 2023Q1-2024Q2 6 rows Select algorithm and tune hyperparameters.
TEST 2024Q3-2026Q2 8 rows Locked holdout; evaluated after selection.

Predictor definitions (one-quarter lag)

lag1_pat_margin_pct: prior reported quarter PAT margin.
lag1_ebitda_margin_pct: prior reported quarter EBITDA margin.
lag1_usd_inr_q_mean: prior-quarter mean USD/INR.
lag1_cpi_inflation_q_mean: prior-quarter mean CPI inflation.
lag1_wpi_inflation_q_mean: prior-quarter mean WPI inflation.
lag1_brent_monthly_q_mean: prior-quarter mean monthly Brent benchmark.

Leakage controls and limitation

All predictors are shifted from the immediately preceding reported quarter; target-quarter
PAT margin is label only. Gaps are not bridged. Test data are not used for tuning. Ridge imputation and scaling are fitted inside the training pipeline; Random Forest uses median imputation without scaling.
Limitation: the workbook has no historical publication timestamps. Lagging prevents same-quarter target leakage but cannot prove that every input was public at the exact historical forecast date. This is a chronological academic backtest, not a vintage-controlled backtest. Calendar-quarter labels are retained as supplied.

Baseline and candidate experiments

3. Baselines first

Primary metric: mean absolute error (MAE), in PAT-margin percentage points.
Secondary metric: root mean squared error (RMSE), which penalises large misses.
Persistence rule: predict margin(t) = margin(t-1).
Validation baseline: MAE 12.18 pp | RMSE 13.92 pp.

Candidate models and tuning approach

Ridge Regression: alpha in {0.1, 1, 10, 100}; median imputation, StandardScaler and Ridge. Best validation setting: alpha=100; MAE 42.62 pp; RMSE 43.94 pp.
Random Forest: 300 trees; max_depth in {2,3,4}; min_samples_leaf in {2,3}; max_features=1.0; random_state=42. Best setting: depth=2, leaf=3; validation MAE 51.95 pp; RMSE 54.15 pp.

Validation comparison

Persistence baseline MAE 12.18 pp RMSE 13.92 pp
Ridge (alpha=100) MAE 42.62 pp RMSE 43.94 pp
Random Forest (depth=2, leaf=3) MAE 51.95 pp RMSE 54.15 pp
Both candidate families underperform persistence on the six-quarter validation window. The current data do not show stable incremental signal beyond recent PAT margin. Ten candidate configurations plus baseline/final test are logged: 12 runs.

Locked test and reproducibility

4. Locked test result

After selection, persistence was evaluated once on the untouched 2024Q3-2026Q2 test.
Selected model: Persistence baseline
Test MAE: 11.53 pp | Test RMSE: 14.20 pp | MAE improvement vs baseline: 0.0%.
The average absolute error is material relative to ordinary positive margins. This is a benchmark result, not evidence of production-ready forecasting accuracy.

Reproducibility package

src/d4_train.py: supervised data preparation, fixed split, baseline, model tuning, locked holdout evaluation and run logging.
notebooks/D4_modeling_experiment.ipynb: notebook entry point and result inspection.
experiments/D4_experiment_log.csv: date, data hash, features, parameters, seed, metric.
D4_candidate_results.csv, D4_metrics.json and D4_test_predictions.csv: results.
D4_data_manifest.json: source file, worksheet, row accounting and SHA-256.
D4_environment.json and requirements-d4.txt: runtime and pinned dependencies.
tests/test_d4_train.py: lag and split tests (2 passed locally).

Run instructions

1. Create and activate a Python virtual environment.
2. pip install -r requirements-d4.txt
3. From repository root: python -m src.d4_train
4. Test: pytest -q tests/test_d4_train.py

The script defaults to data/processed/Final_Quarterly_Data(1).xlsx.

Reproducibility identifiers and limitations

Run date: 2026-11-07 | Random seed: 42
Source SHA-256: 85bf9e7ee51860fa38892ba3cb3e24b74887645bb2570c096f6183e59a97670c
Only 35 supervised rows; calendar-quarter labels retained; historical publication timestamps absent; Brent is a crude benchmark proxy, not IndiGo-specific ATF expense.
Extreme loss quarters are retained, not winsorised. D5 should add vintage controls, regime error analysis, rolling-origin backtests and forecast uncertainty.
Linked course outcome: CO3.